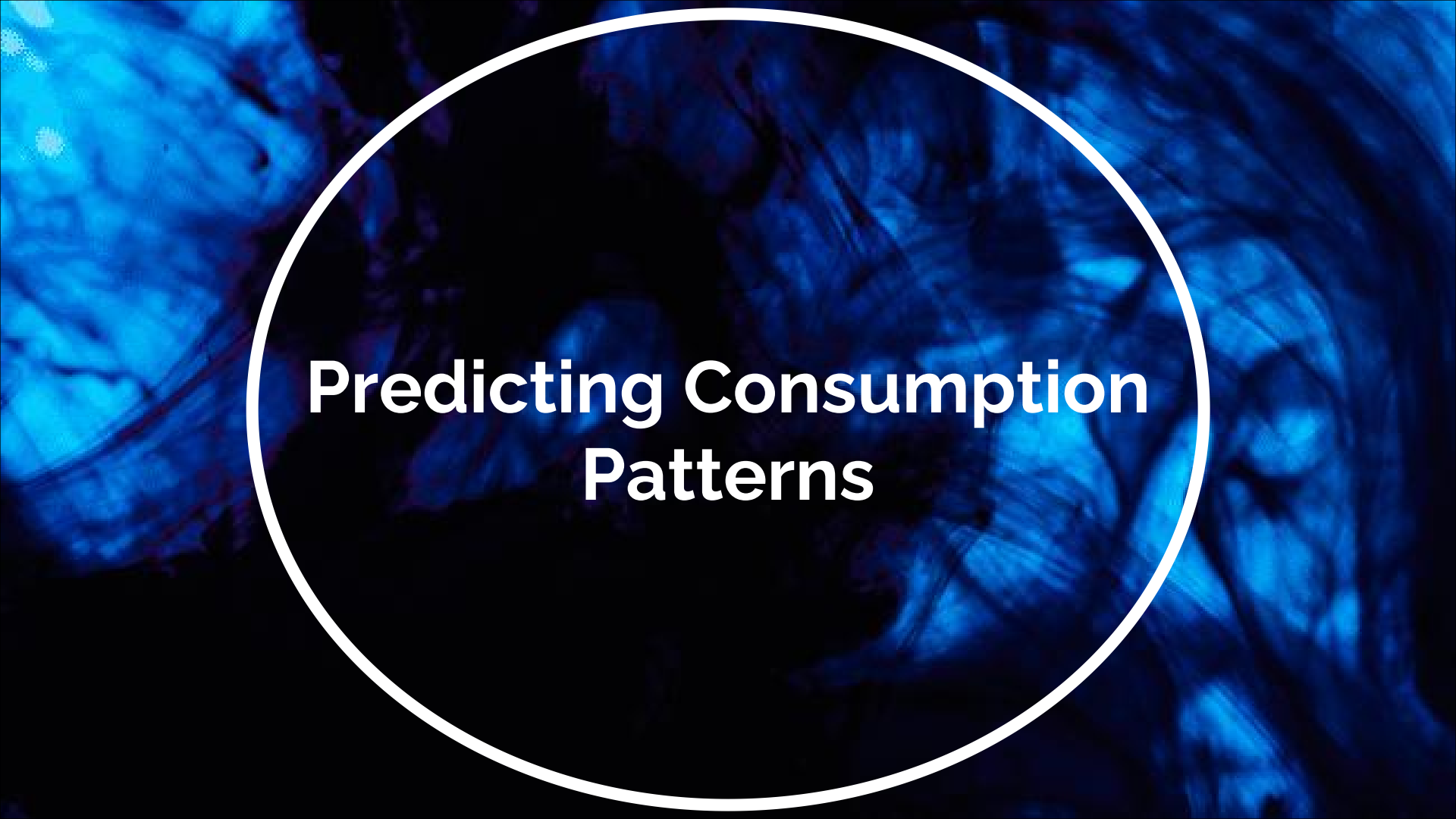


The background is a dark blue, textured surface with swirling, organic patterns. A large, thin white circle is centered on the image, framing the text.

Predicting Consumption Patterns

TABLE OF CONTENTS

INTRODUCTION	01	04	DATASETS
LITERATURE SURVEY	02	05	STEPS
INFERENCE	03	06	POSSIBLE ALGORITHMS



INTRODUCTION

Usually, individuals consume a few items from a large range of potential items. However, there are several circumstances where it is important to make predictions for new items based on previously consumed items for a person, rather than just recommending fresh items. In their ability to capture important information in historical behavior, commonly used matrix factorization methods are constrained, resulting in relatively low predictive precision for these types of problems. Using many real-world datasets, we try to analyze our model in terms of accuracy in user usage predictions. Compared to a number of state-of-the-art matrix factorization methods, the mixture model approach is proved to be systematically more precise and more effective for these issues.



LITERATURE SURVEY

Predicting Consumption Patterns with Repeated and Novel Events-2018

Authors: *Dimitrios Kotzias, Moshe Lichman, and Padhraic Smyth*

Abstract: *This paper shows that the mixture model approach is systematically more accurate and more efficient for the problem compared to a variety of state-of-the-art matrix factorization methods.*

Algorithm/Method: *Maximum Likelihood and Bayesian Estimation, Learning predictive mixture weights, Parameter estimation*

Future Work: *Future directions include leveraging temporal information in a scalable manner for additional improvements in predictive accuracy*



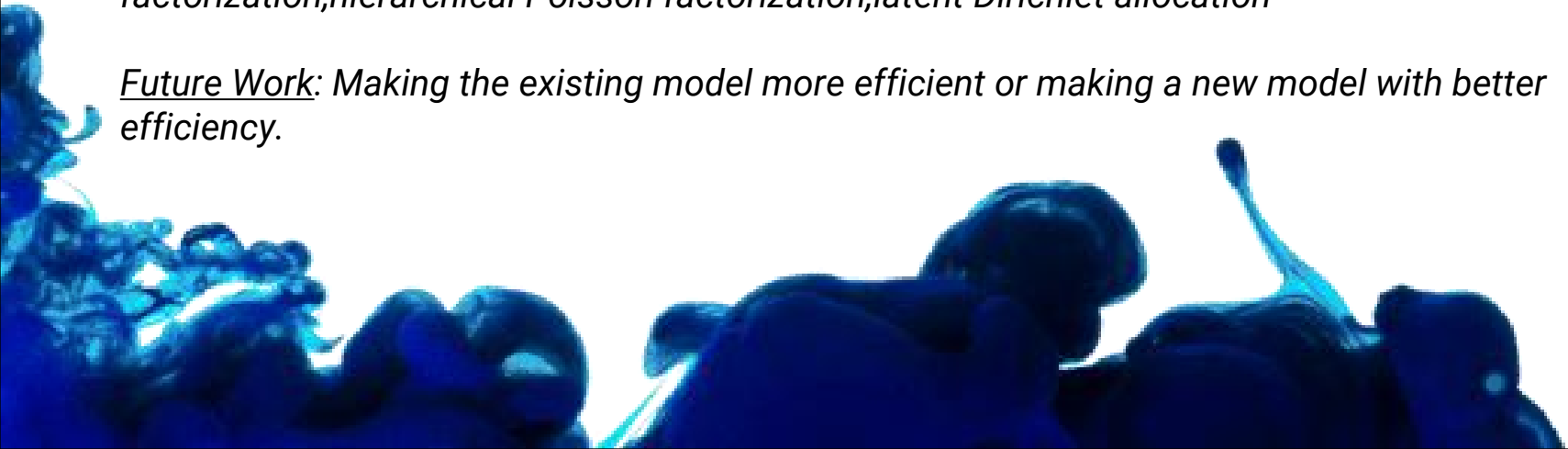
Characterizing and Predicting Repeat Food Consumption Behavior for Just-in-Time Interventions

Authors: Yue Liu, Helena Lee, Palakorn Achananuparp, Ee-Peng Lim, Tzu-Ling Cheng, Shou-De Lin

Abstract: This paper shows that algorithms incorporating the exploration and exploitation and temporal dynamics are more effective in the next day recommendation task than most state-of-the-art algorithms.

Algorithm/Method: MixtureTW, Factorizing Personalized Markov Chains, non-negative matrix factorization, hierarchical Poisson factorization, latent Dirichlet allocation

Future Work: Making the existing model more efficient or making a new model with better efficiency.



The Dynamics of Repeat Consumption

Authors: Ravi Kumar, Andrew Tomkins, Sergei Vassilvitskii, Ashton Anderson

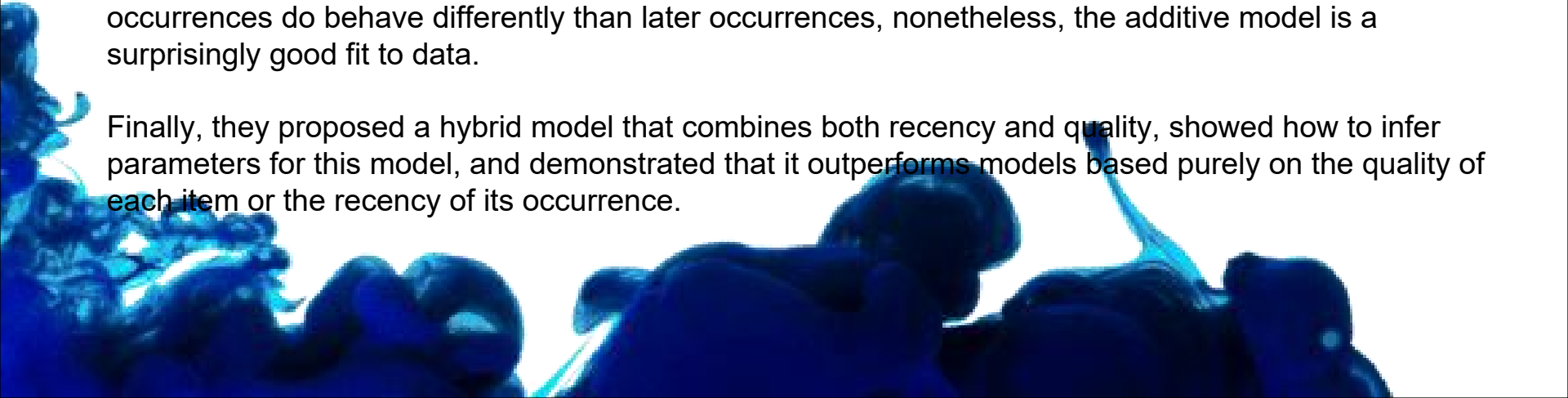
Study: They studied the dynamics of repeat consumption over a range of datasets, and showed a number of consistent patterns.

First, they developed an additive model of recency, and showed how to infer parameters for this model.

Second, they showed that over a range of datasets, the inferred values are well-fit by a power law with exponential cutoff, requiring only three parameters, and producing high likelihood of observed data.

Third, they explored the limitations of their additive model of recency, and showed that while first occurrences do behave differently than later occurrences, nonetheless, the additive model is a surprisingly good fit to data.

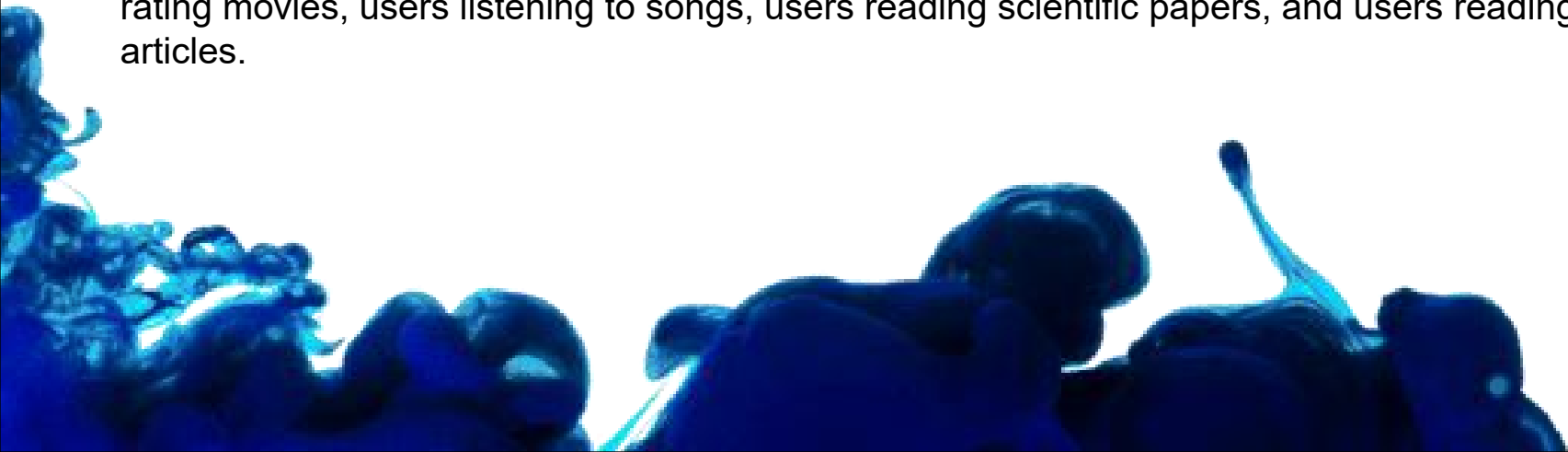
Finally, they proposed a hybrid model that combines both recency and quality, showed how to infer parameters for this model, and demonstrated that it outperforms models based purely on the quality of each item or the recency of its occurrence.



Scalable Recommendation with Poisson Factorization

Authors: Prem Gopalan, Jake M. Hofman, David M. Ble

Summary: They developed hierarchical Poisson matrix factorization (HPF) for recommendation. HPF models sparse user behavior data, large user/item matrices where each user has provided feedback on only a small subset of items. HPF handles both explicit ratings, such as a number of stars, or implicit ratings, such as views, clicks, or purchases. They developed a variational algorithm for approximate posterior inference that scales up to massive data sets, and we demonstrate its performance on a wide variety of real-world recommendation problems—users rating movies, users listening to songs, users reading scientific papers, and users reading news articles.



SLIM: Sparse Linear Methods for Top-N Recommender Systems -2011

Authors: Xia Ning and George Karypis

Abstract: This paper focuses on developing effective and efficient algorithms for top-N recommender systems.

Algorithm/Method: Sparse Linear Method (SLIM) is proposed, which generates top N recommendations by aggregating from user purchase/rating profiles.

Future Work: Making the existing model more efficient or making a new model with better efficiency.



A survey of collaborative filtering based social recommender systems

Authors: Xiwang Yang, Yang Guo b, Yong Liu a, Harald Steck

Abstract: In this paper, a survey of CF(collaborative filtering)-based social recommender systems were presented.

Algorithm/Method: We classify CF-based social recommender systems into two categories: matrix factorization based social recommendation approaches and neighborhood based social recommendation approaches. Both types of approaches are surveyed and compared.

Future Work: Making the existing model more efficient or making a new model with better efficiency.



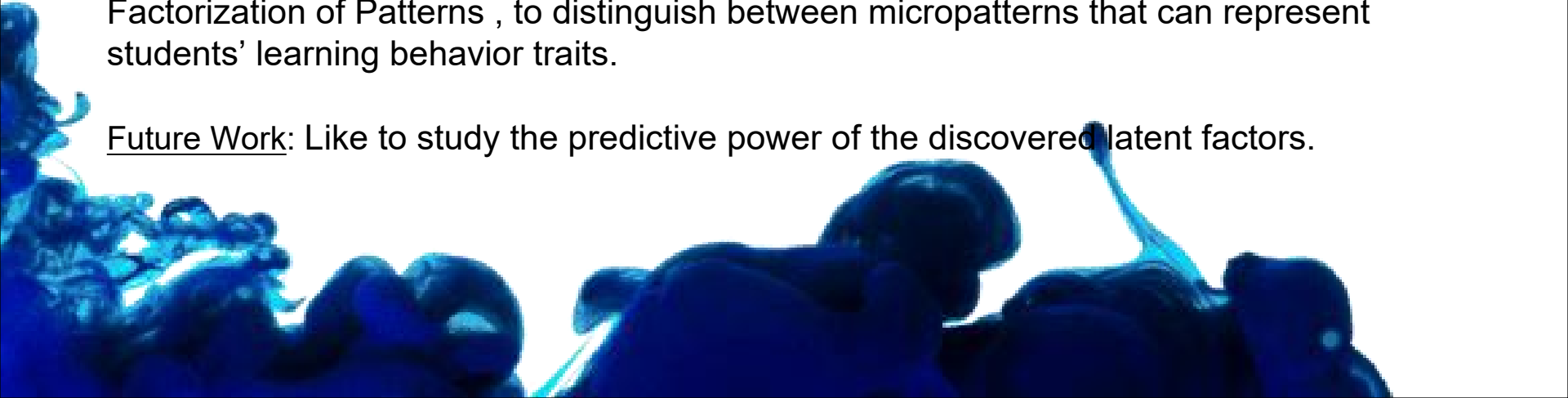
Detecting Trait versus Performance Student Behavioral Patterns Using Discriminative Non-Negative Matrix Factorization

Authors: Mehrdad Mirzaei, Shaghayegh Sahebi, Peter Brusilovsky

Abstract: In this paper, a survey of students follow stable behavioral patterns while learning in online educational systems.

Algorithm/Method: At first CM-SPAM, a sequential pattern mining algorithm, to find the frequent micro-patterns with minimum support of 5.4%. Discriminative Non-negative Factorization of Patterns, to distinguish between micropatterns that can represent students' learning behavior traits.

Future Work: Like to study the predictive power of the discovered latent factors.

An abstract graphic at the bottom of the slide featuring a vibrant blue ink splash or smoke effect against a white background. The splash originates from the left side and spreads towards the right, with various swirling and billowing shapes. The colors range from deep navy blue to a lighter, almost white-blue at the edges, creating a dynamic and artistic visual element.

PREDICTION OF CANCER MICROARRAY AND DNA METHYLATION DATA USING NON-NEGATIVE MATRIX FACTORIZATION

Authors: Parth Patel , Kalpdrum Passi and Chakresh Kumar Jain.

Abstract: This paper , suggests that different algorithms and approaches for the reduction of dimensionality of such microarray datasets.

Algorithm/Method: A Popular feature reduction techniques include Principal Component Analysis (PCA), Non-negative Matrix Factorization (NMF), Linear Discriminant Analysis (LDA) is used in this project.

Future Work: Use of NMF for methylation datasets is limited in the context that a lower accuracy is observed for these cases. Further studies aims on increase in accuracy.



INFERENCE

We separate related work into three broad categories:

- 1. Models that focus on discovering new items*
- 2. Models that focus on re-consumption of old items*
- 3. Models that predict a mix of new and repeat consumption.*

Our work belongs in the last category as it focuses on balancing the two primary driving forces of human behavior in this context—discovery and reconsumption

In this paper, as well as investigating individual weights and broader classes of datasets, we provide a significantly more in-depth analysis of explore/exploit behavior in the context of user-item consumption.



DATASET

(User x Items)

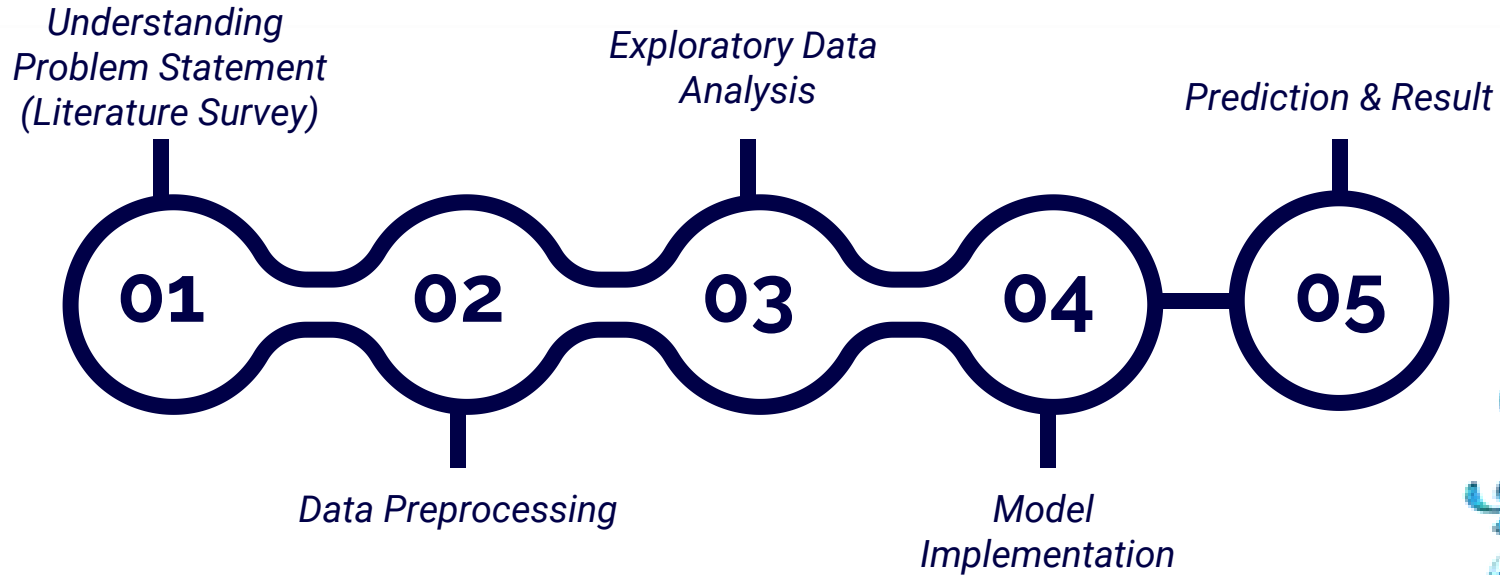
lastfm (992 x 15k): How many times, a user listened to each artist. Covers 3 years of listening habits.

- Dataset has 3 columns(user id,item id,frequency)

LINK: <https://archive.ics.uci.edu/ml/machine-learning-databases/00441/>



STEPS



WE ARE..

Abhay P (001)

Sai Akhil Y (050)

Vijay Mani Dev K (031)

Aravind T (062)



Machine Learning and Data Mining

- 15CSE402

Group-1

Name	Roll No.
Abhay P	CB.EN.U4CSE17001
Kaluri Vijay Mani Dev	CB.EN.U4CSE17031
Sai Akhil Y	CB.EN.U4CSE17050
Aravind T	CB.EN.U4CSE17062

1. Introduction

Usually, individuals consume a few items from a large range of potential items. However, there are several circumstances where it is important to make predictions for new items based on previously consumed items for a person, rather than just recommending fresh items. In their ability to capture important information in historical behavior, matrix factorisation is a commonly used technique.

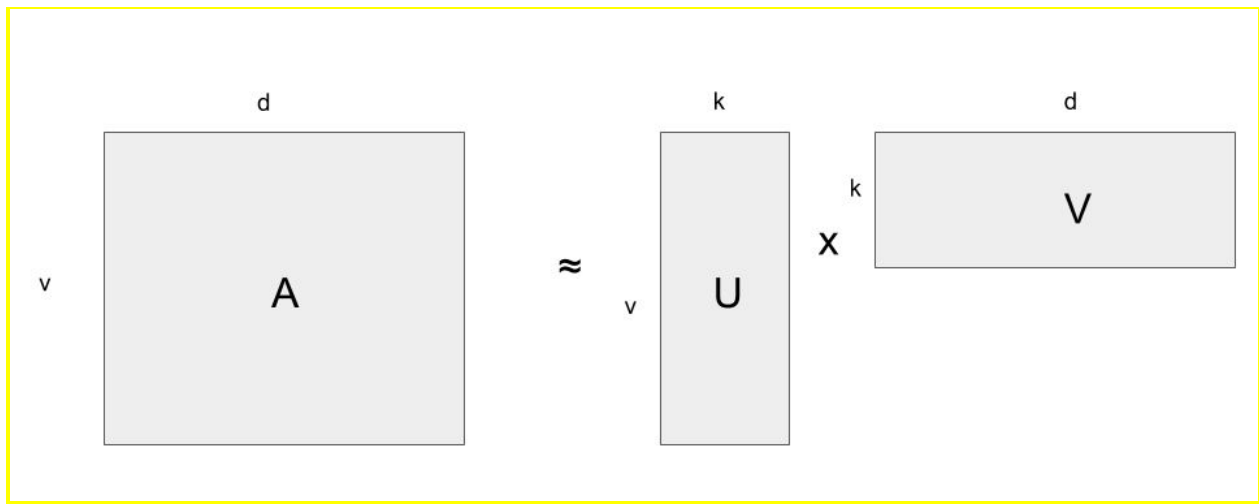
2. Objective

To build a model to predict user-song compatibility using matrix factorisation and cosine similarity methods and is analyzed in terms of loss calculation.

3. Problem statement and Assumption

Predicting Consumption Patterns with Repeated and Novel Events

4. Architecture



Matrix factorization algorithms work by decomposing the original matrix into two matrices one is the upper triangle (U), and the other is the lower triangle (L).

In matrix factorization, we basically break a matrix into usually 2 smaller matrices each with smaller dimensions. These matrices are often called 'Embeddings'. We can have variants of Matrix Factorization-> 'Low Rank MF' , 'Non-Negative MF' (NMF) and so on.

Here, 'Low Rank Matrix Factorization' is being used. Embeddings are created for both users as well as the item; song in our case. The number of dimensions or the so called 'Latent Factors' in the embeddings is a hyperparameter to deal with in this implementation of Collaborative Filtering.

5. Related work

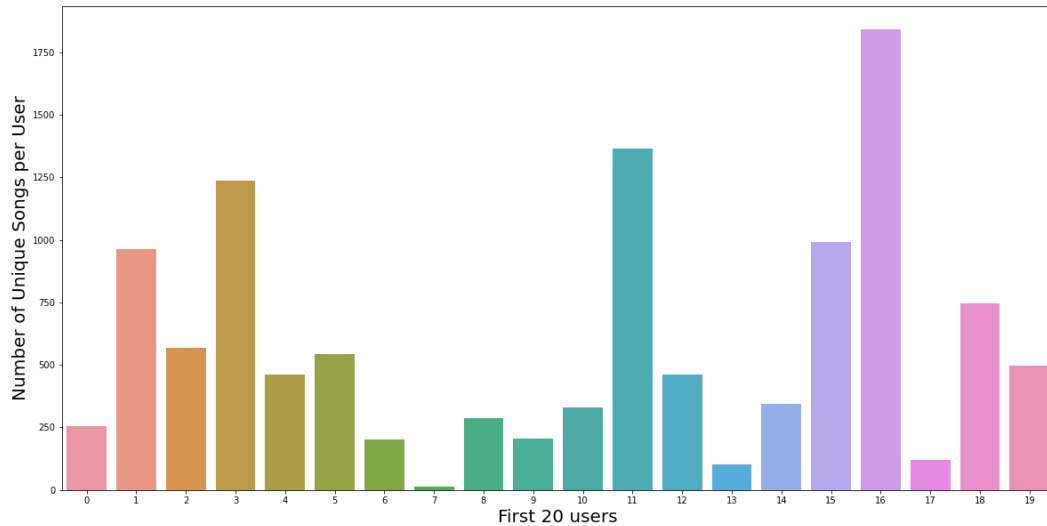
Matrix factorization techniques are widely used in research on recommender systems. In such applications, the model is primarily tasked with learning user preferences from historical observations and predicting future user-item interactions with a focus on the discovery of new items [5], [6], [7].

6. Data collection and preparation

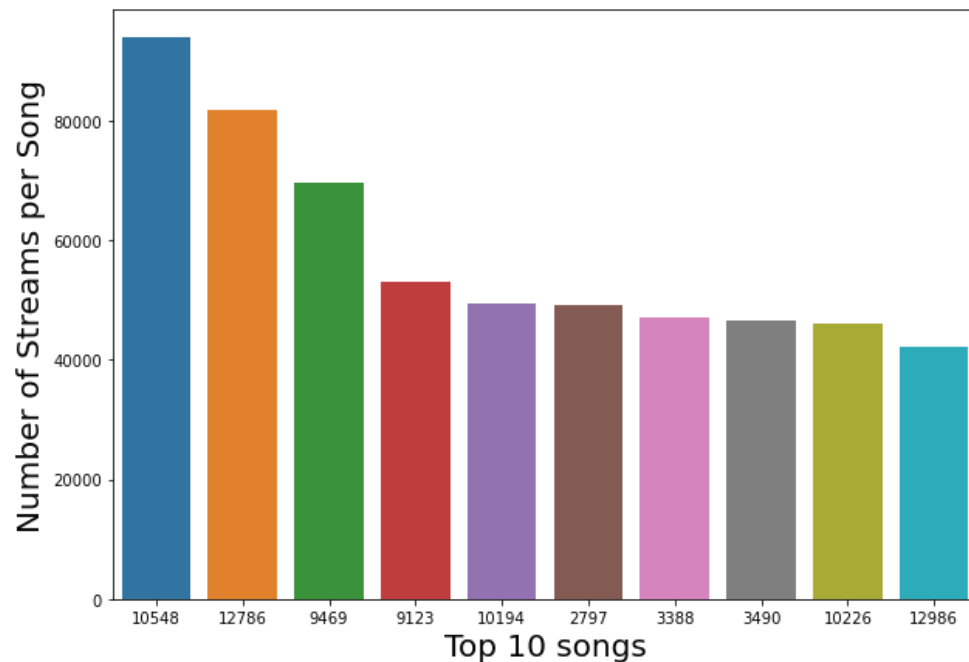
a. Feature Selection and/or Feature Extraction

We also computed the correlation, since there are limited features in the dataset, feature extraction does not seem feasible.

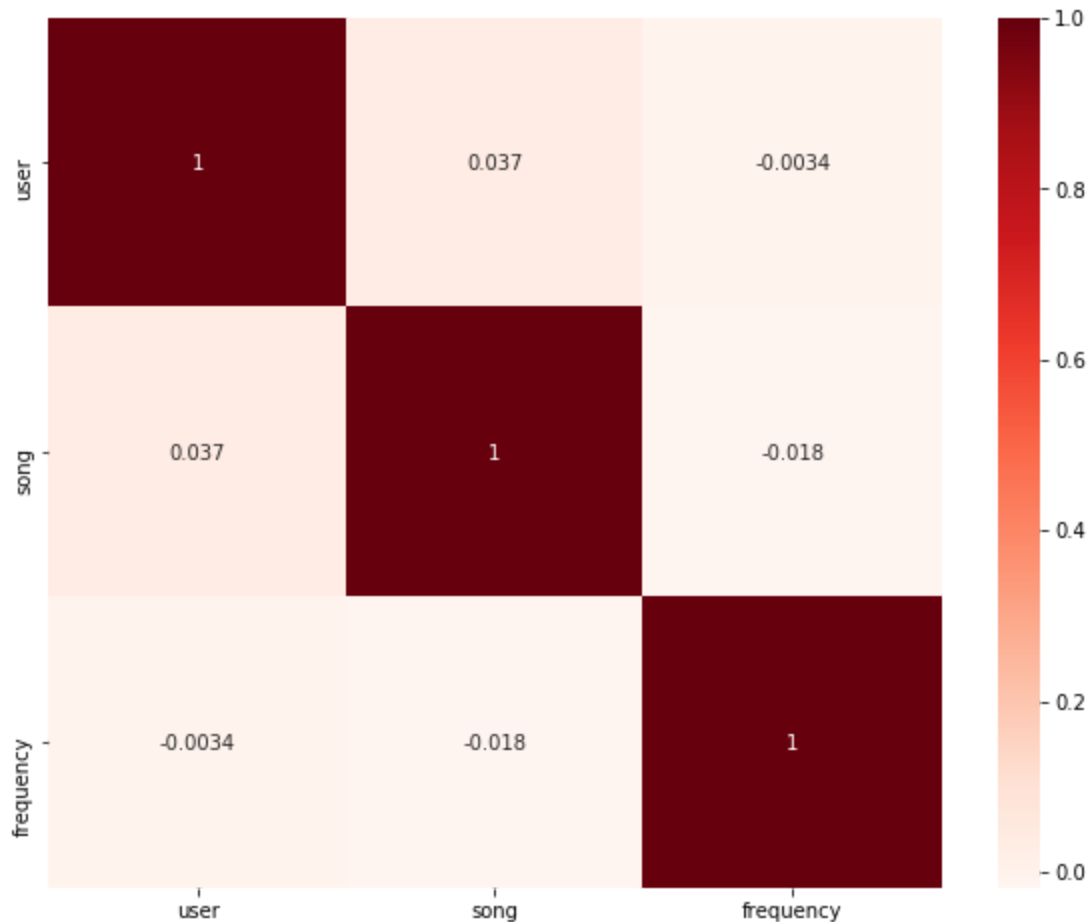
b. Data Visualization and Hypothesis



The above plot is bar plot where we can see the users with different mindsets - from user listening to selected number of songs to users listening to wide variety of songs



Here we can see the top 10 songs streamed by users



Here we can see the correlation between each pair of columns.

c. Data Cleaning & Pre-Processing

We used a clean dataset that had columns(user,song,frequency). No preprocessing was required for the dataset

7. Data Modeling and Inference

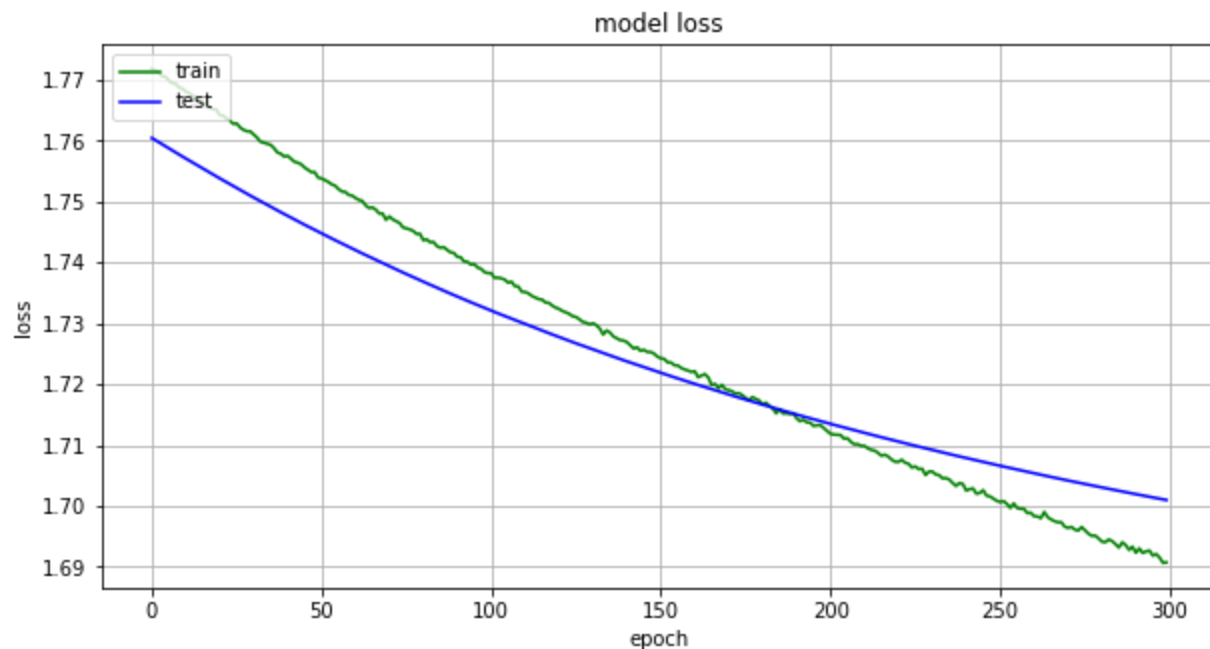
a. Model

For the first model embeddings were used for each user and song using embeddings function from keras.layers and later embeddings were flattened from matrix form to 1-d array. Later dot product of both user and song vectors is calculated and created a model using it. We used Adam optimiser to optimize our model. Then model is fitted and was run for 300 epochs and plotted the graph between training loss and testing loss

For the second model cosine similarity on both users and songs is taken. Through this model prediction of similar songs to a particular song, prediction of similar users (like similar mindsets). A function which predicts the frequency given user and song. MAE between actual and predicted values for few users is calculated

b. Performance Evaluation and result discussion

The loss has been computed with respect to every epoch and a graph has been plotted about the same shown below. We can see that as epochs increase the training loss and validation loss tends to converge. We observe different epoch vs loss graphs with changes in dropout value and hyper parameter value.



b. Model Tweaking, Regularization, Hyper Parameter Tuning

We have used the Dropout function to avoid overfitting of the model. In our model when we are converting user and song inputs into embeddings with the help of keras. We have a hyperparameter called `n_latent_factors` which helps us in determining the dimensions of our embeddings. The values were tuned so as to reduce the difference between training loss and validation loss.

8. Conclusion and Future Enhancements

Using a real-world dataset, the user-song compatibility was computed using matrix factorisation and the loss was analysed and cosine similarity was computed to group the similar songs or users or predict frequencies.

In future, we will try making the existing model more efficient or making a new model with better efficiency.

9. References:

- [1]D. Kotzias, M. Lichman and P. Smyth, "Predicting Consumption Patterns with Repeated and Novel Events," in *IEEE Transactions on Knowledge and Data Engineering*, vol. 31, no. 2, pp. 371-384, 1 Feb. 2019, doi: 10.1109/TKDE.2018.2832132.
- [2] Liu, Yue et al. "Characterizing and Predicting Repeat Food Consumption Behavior for Just-in-Time Interventions". *Proceedings of the 9th International Conference on Digital Public Health - DPH2019*. (2019).
- [3]Xiwang Yang, Yang Guo, Yong Liu, & Harald Steck (2014). A survey of collaborative filtering based social recommender systems*Computer Communications*, 41, 1 - 10.
- [4]X. Ning and G. Karypis, "SLIM: Sparse Linear Methods for Top-N Recommender Systems," *2011 IEEE 11th International Conference on Data Mining*, Vancouver,BC,
- [5] X. Yang, Y. Guo, Y. Liu, and H. Steck, "A survey of collaborative filtering based social recommender systems," *Comput. Commun.*, vol. 41, pp. 1–10, Mar. 2012011, pp. 497-506, doi: 10.1109/ICDM.2011.134.
- [6]] X. Ning and G. Karypis, "Slim: Sparse linear methods for top-n recommender systems," in *Proceedings of the 11th IEEE International Conference on Data Mining*, ser. ICDM '11. IEEE, 2011, pp. 497–506.

[7] D. Kotkov, S. Wang, and J. Veijalainen, "A survey of serendipity in recommender systems," *Know.-Based Syst.*, vol. 111, no. C, pp. 180–192, Nov. 2016